

Theta Closure in Modal Triplet Theory II: Direct Geometric Realization of Selected Gauge-Profile Overlaps

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Abstract

We re-execute the Route A geometric overlap ansatz against the selected multi-loop gauge-profile targets of Paper I. In its stated dimensionless normalization, the round- S^2 effective lens base and isotropic compact Heisenberg nilmanifold can be retargeted analytically to

$$(f_2 R_{\text{lens}})^2 = 0.2555137 R_1, \quad c = 0.9948493 R_1.$$

The associated spectral lower bounds remain above the assumed dimensionless admissibility floor. This proves existence of a calibrated representative in the declared auxiliary ansatz. It does not prove that the full lens space reduces to this S^2 model, identify the Lens–Nil representative with the selected q79/Fu–Yau compactification, select a geometry uniquely by MTT, or make the gauge couplings held-out predictions. The shared circle is common $U(1)$ phase/holonomy data, not an additional factor in a $S^1 \times L(3, 1) \times \text{Nil}_3$ manifold. The old 5 TeV target pair is withdrawn.

Revision note for this edition

Supersedes. *Theta Closure in Modal Triplet Theory II: Direct Geometric Realization of Non-abelian Overlaps*, first edition.

Reason. The previous numerical target came from the retired few-TeV crossing and the auxiliary Lens–Nil ansatz was liable to be read as the selected q79 compactification or a literal circle product.

Resolution. Version 2 retargets the analytic ansatz to the Paper I SMDR profile, counts the shared circle only as common holonomy data, and labels the round- S^2 /nil model auxiliary and calibrated.

Retained result. An explicit representative satisfying the two target ratios and dimensionless spectral-floor test exists in the declared ansatz.

Remaining boundary. No uniqueness, full lens reduction, q79/Fu–Yau identification, or held-out gauge prediction is established.

1 Relation to the revised Θ –closure core paper

Paper I now transports measured source coordinates with SMDR v1.3 to the full-Standard-Model $\overline{\text{MS}}$ profile at $Q = M_t$. In the same GUT-normalized overlap convention it supplies

$$\frac{I_2}{I_1} = 0.5110273 \pm 0.0001231, \quad \frac{I_3}{I_1} = 0.158335 \pm 0.001098.$$

The scale $Q = M_t$ is a matching convention, not an internal gap or coherence scale. This paper asks only whether the explicit Route A ansatz contains a representative with those overlap ratios

and the stated dimensionless gap margin. Because the gauge rows select the target, success is a calibrated realization and round-trip test.

2 Overlap definitions

We recall the effective gauge-kinetic coefficients defined in Paper I:

$$I_a = \int_{B_a} w_a \langle \omega_a, \omega_a \rangle d\mu_{B_a}, \quad a = 1, 2, 3,$$

where B_a denotes a sector support after spectator integrations. The three supports are not asserted to be simultaneous Cartesian factors of one compactification. The same shared-circle phase datum is reused in all lanes and counted once.

In the auxiliary coefficient ansatz, the abelian coefficient I_1 is fixed by the declared shared-circle normalization:

$$I_1 = 2\pi R_1.$$

We compute $I_2^{(0)}$ and $I_3^{(0)}$ below.

3 Lens sector: computation of $I_2^{(0)}$

Dimensional and geometric convention. All radii and overlaps below are dimensionless after one common internal length unit has been factored out. The “lens layer” used in the calculation is an effective two-dimensional constant-curvature base modeled by a round S^2 ; it is not the full three-dimensional lens space $L(3, 1)$. A full lens-space realization requires a separate reduction theorem.

3.1 Lens geometry

The declared auxiliary lens-base ansatz uses the spectral bound

$$\lambda_{\text{lens}} \geq \frac{2}{(f_2 R_{\text{lens}})^2}.$$

Remark 3.1 (Why the round S^2 model is used here). The coefficient $2/R^2$ is the exact first nonzero Laplace–Beltrami eigenvalue on the round two-sphere of radius R . Thus modeling the lens layer as a constant-curvature S^2 scaled by the effective radius $f_2 R_{\text{lens}}$ supplies one explicit geometry with the stated spectral bound. It is not selected uniquely by that eigenvalue.

The coefficient “2” motivates the round two-sphere as the baseline effective model within this ansatz; it is not a global uniqueness theorem. We take

$$\Sigma_2 \simeq S^2, \quad g_{\Sigma_2} = (f_2 R_{\text{lens}})^2 g_{S^2}.$$

3.2 Lens harmonic

The massless gauge representative is taken to have unit pointwise norm on Σ_2 after gauge fixing. The shared-circle phase is a common spectator whose normalization is absorbed into Paper I’s coefficient convention. With the auxiliary weight $w_2^{(0)} = 1$, this gives

$$I_2^{(0)} = \text{Area}(\Sigma_2) = 4\pi (f_2 R_{\text{lens}})^2.$$

3.3 Matching to Θ -target

From Paper I,

$$\frac{I_2}{I_1} \approx 0.5110273, \quad I_1 = 2\pi R_1.$$

Hence

$$\frac{4\pi(f_2 R_{\text{lens}})^2}{2\pi R_1} = 0.5110273 \quad \Rightarrow \quad (f_2 R_{\text{lens}})^2 = 0.2555137 R_1.$$

Proposition 3.2 (Lens overlap closure). *The lens overlap $I_2^{(0)}$ matches the Θ -target for $(f_2 R_{\text{lens}})^2 = 0.2555137 R_1$.*

3.4 Lens admissibility

The spectral bound gives

$$\lambda_{\text{lens}} \geq \frac{2}{0.2555137 R_1}.$$

Since admissibility of the circle sector requires $R_1 \leq 2$, we obtain $\lambda_{\text{lens}} \geq 3.57 \gg 0.25$. Thus the lens sector is safely admissible.

4 Nil sector: computation of $I_3^{(0)}$

4.1 Nil geometry

We take Σ_3 to be the standard compact Heisenberg nilmanifold $\Gamma \backslash \text{Nil}_3$ with integer lattice and left-invariant metric

$$g_{\text{nil}} = a^2 \sigma_1^2 + b^2 \sigma_2^2 + c^2 \sigma_3^2,$$

where

$$\sigma_1 = dx, \quad \sigma_2 = dy, \quad \sigma_3 = dz - x dy.$$

4.2 Canonical massless harmonics

The first cohomology is generated by the closed forms dx and dy . We choose the isotropic convention

$$a = b,$$

which renders the overlap independent of the basis in $\text{span}\{dx, dy\}$.

4.3 Overlap computation

For $\omega^{(0)} = dx$ or dy , auxiliary weight $w_3^{(0)} = 1$, and the unit coordinate fundamental domain,

$$|\omega^{(0)}|^2 = \frac{1}{a^2}, \quad d\text{vol} = a^2 c (\sigma_1 \wedge \sigma_2 \wedge \sigma_3).$$

Hence

$$I_3^{(0)} = \int |\omega^{(0)}|^2 d\text{vol} = c.$$

4.4 Matching to Θ -target

From Paper I,

$$\frac{I_3}{I_1} \approx 0.158335 \quad \Rightarrow \quad I_3 \approx 0.9948493 R_1.$$

Therefore

$$\boxed{c = 0.9948493 R_1.}$$

Proposition 4.1 (Nil overlap closure). *For the isotropic nil metric $a = b$ with $c = 0.9948493 R_1$, the leading-order nil overlap $I_3^{(0)}$ matches the Θ -target.*

4.5 Nil admissibility

The scalar Laplacian on $\Gamma \backslash \text{Nil}_3$ has spectrum

$$\lambda_1 \geq \min \left\{ \frac{4\pi^2}{a^2}, \frac{2\pi}{a^2} + \frac{4\pi^2}{c^2} \right\}.$$

With $a = b = 1$ and $c \leq 1.989699$ (from $R_1 \leq 2$),

$$\lambda_1 \gtrsim 11 \gg 0.25.$$

Thus the nil sector satisfies the MTT admissibility bound $\lambda_{\text{nil}} \geq 0.25$ with large margin.

5 Route-A result and scope

Within the declared dimensionless ansatz, the selected profile targets admit an explicit lens-base/nil representative and the assumed spectral inequalities remain satisfied. This closes the algebraic retargeting and ansatz-level existence check. It does not close source selection of the geometry, a literal $L(3, 1)$ reduction, global HYM connection data, or a held-out gauge-coupling prediction. In particular, this auxiliary representative is not the q79 Fu–Yau compactification and supplies no theorem identifying the two spaces.

A Spectral lower bound on $\Gamma \backslash \text{Nil}_3$ for the left-invariant metric

This appendix justifies the spectral estimate used in the nil admissibility check.

A.1 Heisenberg nilmanifold and left-invariant metric

Let Nil_3 be the Heisenberg group with global coordinates $(x, y, z) \in \mathbb{R}^3$ and left-invariant 1-forms

$$\sigma_1 = dx, \quad \sigma_2 = dy, \quad \sigma_3 = dz - x dy,$$

satisfying $d\sigma_1 = d\sigma_2 = 0$ and $d\sigma_3 = -\sigma_1 \wedge \sigma_2$.

Let Γ be the standard integer lattice so $\Gamma \backslash \text{Nil}_3$ is compact. Equip it with the left-invariant metric

$$g = a^2 \sigma_1^2 + b^2 \sigma_2^2 + c^2 \sigma_3^2, \quad a, b, c > 0.$$

The dual left-invariant vector fields are

$$X = \partial_x, \quad Y = \partial_y + x \partial_z, \quad Z = \partial_z,$$

with $[X, Y] = Z$ and other brackets zero. An orthonormal frame is

$$e_1 = \frac{1}{a} X, \quad e_2 = \frac{1}{b} Y, \quad e_3 = \frac{1}{c} Z.$$

A.2 Scalar Laplacian

For a left-invariant orthonormal frame, the scalar Laplacian on functions may be written as

$$\Delta f = -(e_1^2 + e_2^2 + e_3^2)f$$

(up to sign convention; we use the nonnegative operator $-\sum e_i^2$). Substituting the frame gives

$$\Delta = -\frac{1}{a^2}\partial_x^2 - \frac{1}{b^2}(\partial_y + x\partial_z)^2 - \frac{1}{c^2}\partial_z^2. \quad (\text{A.1})$$

A.3 Fourier decomposition in the central direction

Because z is periodic on $\Gamma \backslash \text{Nil}_3$, expand

$$f(x, y, z) = \sum_{p \in \mathbb{Z}} e^{2\pi i p z} g_p(x, y).$$

On the p -th mode, $\partial_z \mapsto 2\pi i p$ and (A.1) becomes

$$\Delta_p = -\frac{1}{a^2}\partial_x^2 - \frac{1}{b^2}(\partial_y + 2\pi i p x)^2 + \frac{(2\pi p)^2}{c^2}. \quad (\text{A.2})$$

We bound the lowest nonzero eigenvalue by analyzing $p = 0$ and $p \neq 0$ separately.

A.4 The $p = 0$ sector (torus bound)

For $p = 0$,

$$\Delta_0 = -\frac{1}{a^2}\partial_x^2 - \frac{1}{b^2}\partial_y^2$$

on the flat 2-torus $(x, y) \in (\mathbb{R}/\mathbb{Z})^2$. Its eigenvalues are

$$\lambda_{m,n}^{(0)} = 4\pi^2 \left(\frac{m^2}{a^2} + \frac{n^2}{b^2} \right), \quad (m, n) \in \mathbb{Z}^2.$$

Hence the smallest nonzero eigenvalue in this sector satisfies

$$\lambda_{\min}^{(0)} \geq \frac{4\pi^2}{\max(a^2, b^2)}. \quad (\text{A.3})$$

A.5 The $p \neq 0$ sector (Landau/oscillator lower bound)

For $p \neq 0$, the operator (A.2) is the magnetic Laplacian on a 2-torus with constant magnetic field proportional to p , plus a strictly positive shift $(2\pi p)^2/c^2$. A conservative lower bound follows by dropping the nonnegative x -dependence and using the lowest Landau-level scale:

$$\lambda_{\min}^{(p)} \geq \frac{2\pi|p|}{ab} + \frac{(2\pi p)^2}{c^2}, \quad p \neq 0. \quad (\text{A.4})$$

(For our purposes it suffices to retain the $|p| = 1$ bound.)

Under the isotropic convention $a = b$, this yields

$$\lambda_{\min}^{(1)} \geq \frac{2\pi}{a^2} + \frac{4\pi^2}{c^2}. \quad (\text{A.5})$$

A.6 Combined lower bound

Combining (A.3) and (A.5), we obtain

$$\lambda_1(\Delta_{\Gamma \backslash \text{Nil}_3}) \geq \min \left\{ \frac{4\pi^2}{a^2}, \frac{2\pi}{a^2} + \frac{4\pi^2}{c^2} \right\}, \quad (a = b). \quad (\text{A.6})$$

In particular, for $a = b = 1$ and any $c \leq 1.989699$, one has

$$\lambda_1 \geq 2\pi + \frac{4\pi^2}{c^2} \geq 2\pi + \frac{4\pi^2}{(1.989699)^2} \approx 6.283 + 4.77 \approx 11.05,$$

which is much larger than the admissibility floor 0.25 used in Paper I.

Remark A.1. The estimates above are intentionally conservative and suffice only to demonstrate a large spectral margin. A full spectral decomposition of Δ on the compact nilmanifold is known in the literature, but is not required for the present admissibility verification.

References

1. Paper I: *Gauge Couplings, Internal Geometry, and Θ -Closure in MTT*.
2. Standard references on Heisenberg nilmanifolds and spectra (e.g. any textbook reference you prefer).

B Conclusion

The Route A formulas have been evaluated in the same normalization and scheme as revised Paper I. The new central representative is

$$(f_2 R_{\text{lens}})^2 = 0.2555137 R_1, \quad c = 0.9948493 R_1,$$

with uncertainties inherited from the two profile ratios. The nil spectral estimate and lens lower bound continue to exceed the assumed dimensionless floor for the stated parameter range.

The earlier numbers $0.280 R_1$ and $1.439 R_1$ belonged to the withdrawn 5 TeV profile and are not retained. The present construction is a calibrated existence result: the target gauge rows were used to determine the geometry. Selection of this geometry before empirical comparison, literal lens-space reduction, and a global HYM representative remain separate proof obligations.

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